

Please check the examination details below before entering your candidate information

|                                                                                                                                                           |                                                                                                          |                                                                                     |                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------|
| Candidate surname                                                                                                                                         |                                                                                                          | Other names                                                                         |                     |
| <b>Pearson Edexcel</b><br><b>International</b><br><b>Advanced Level</b>                                                                                   | Centre Number                                                                                            | Candidate Number                                                                    |                     |
|                                                                                                                                                           | <input type="text"/> <input type="text"/> <input type="text"/> <input type="text"/> <input type="text"/> | <input type="text"/> <input type="text"/> <input type="text"/> <input type="text"/> |                     |
| <b>Thursday 14 January 2021</b>                                                                                                                           |                                                                                                          |                                                                                     |                     |
| Afternoon (1 hour 30 minutes)                                                                                                                             |                                                                                                          | Paper Reference <b>WCH12/01</b>                                                     |                     |
| <b>Chemistry</b><br><b>International Advanced Subsidiary / Advanced Level</b><br><b>Unit 2: Energetics, Group Chemistry, Halogenoalkanes and Alcohols</b> |                                                                                                          |                                                                                     |                     |
| <b>You must have:</b><br>Scientific calculator, Data booklet, Ruler                                                                                       |                                                                                                          |                                                                                     | Total Marks<br><br> |

### Instructions

- Use **black** ink or **black** ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions.
- Answer the questions in the spaces provided  
– *there may be more space than you need.*
- Show all your working in calculations and include units where appropriate.

### Information

- The total mark for this paper is 80.
- The marks for each question are shown in brackets  
– *use this as a guide as to how much time to spend on each question.*
- In the question marked with an **asterisk (\*)** marks will be awarded for your ability to structure your answer logically, showing how the points that you make are related or follow on from each other where appropriate.
- A Periodic Table is printed on the back cover of this paper.

### Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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**Pearson**

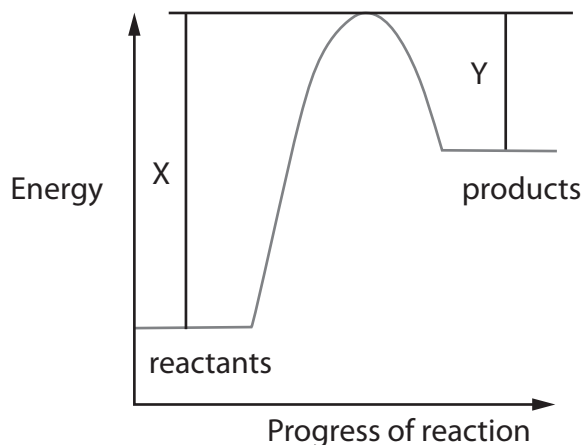
## SECTION A

Answer ALL the questions in this section.

You should aim to spend no more than 20 minutes on this section.

For each question, select one answer from A to D and put a cross ☐. If you change your mind, put a line through the box ☒ and then mark your new answer with a cross ☐.

- 1 The energy profile for a reaction is shown.



What is the minimum energy needed for this reaction to occur?

- ☐ A X  
☐ B Y  
☐ C  $X - Y$   
☐ D  $X + Y$

(Total for Question 1 = 1 mark)

- 2 Which of these isomers with the formula  $C_8H_{18}$  has the lowest boiling temperature?

- ☐ A  $CH_3CH(CH_3)CH_2CH_2CH(CH_3)CH_3$   
☐ B  $CH_3CH(CH_3)CH_2CH_2CH_2CH_2CH_3$   
☐ C  $CH_3C(CH_3)_2CH_2CH(CH_3)CH_3$   
☐ D  $CH_3CH_2CH_2CH_2CH_2CH_2CH_2CH_3$

(Total for Question 2 = 1 mark)



3 Which of these pure compounds has hydrogen bonding in the liquid state?

- ☐ A 1,1,1-trichloroethane,  $\text{CH}_3\text{CCl}_3$
- ☐ B trimethylamine,  $(\text{CH}_3)_3\text{N}$
- ☐ C hydrogen fluoride, HF
- ☐ D hydrogen sulfide,  $\text{H}_2\text{S}$

(Total for Question 3 = 1 mark)

4 The hydrogen ion,  $\text{H}^+$ , bonds to a water molecule forming an  $\text{H}_3\text{O}^+$  ion.

For  $\text{H}_3\text{O}^+$ , what shape and H-O-H bond angle are predicted by the electron-pair repulsion theory?

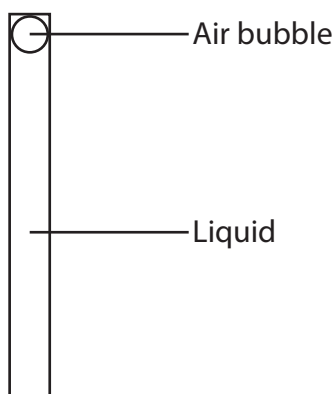
- ☐ A trigonal planar  $120^\circ$
- ☐ B trigonal planar  $117.5^\circ$
- ☐ C trigonal pyramidal  $107^\circ$
- ☐ D trigonal pyramidal  $104.5^\circ$

(Total for Question 4 = 1 mark)

Use this space for any rough working. Anything you write in this space will gain no credit.



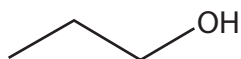
- 5 The viscosities of four liquid organic compounds are compared by placing the liquids in separate identical tubes, each with an air bubble at the top.



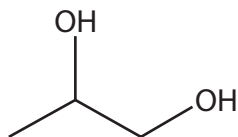
The tubes are inverted, and the times measured for the air bubble to travel the length of the tube.

For which compound does the air bubble take the longest time?

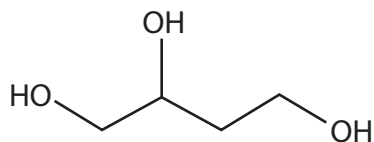
☐ A



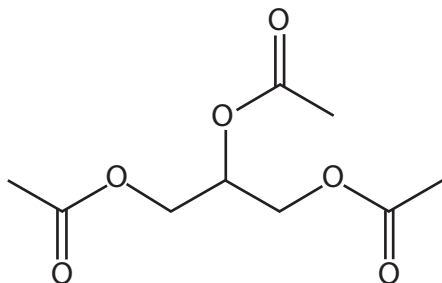
☐ B



☐ C



☐ D



(Total for Question 5 = 1 mark)



6 This question is about the reaction of potassium iodide with concentrated sulfuric acid.

(a) Which of these would **not** be seen?

(1)

- ☐ **A** misty fumes
- ☐ **B** black solid
- ☐ **C** yellow solid
- ☐ **D** dense white smoke

(b) One of the reaction products is the gas  $\text{H}_2\text{S}$ .

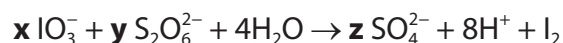
What is the **change** in oxidation number of sulfur as  $\text{H}_2\text{S}$  forms?

(1)

- ☐ **A** -8
- ☐ **B** -6
- ☐ **C** -2
- ☐ **D** +6

(Total for Question 6 = 2 marks)

7 Iodate(V) ions,  $\text{IO}_3^-$ , oxidise dithionate ions,  $\text{S}_2\text{O}_6^{2-}$ , according to the equation



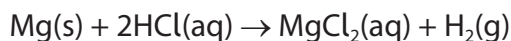
What are the balancing numbers **x**, **y** and **z**?

|                                   | <b>x</b> | <b>y</b> | <b>z</b> |
|-----------------------------------|----------|----------|----------|
| <input type="checkbox"/> <b>A</b> | 2        | 1        | 2        |
| <input type="checkbox"/> <b>B</b> | 2        | 2        | 4        |
| <input type="checkbox"/> <b>C</b> | 2        | 5        | 5        |
| <input type="checkbox"/> <b>D</b> | 2        | 5        | 10       |

(Total for Question 7 = 1 mark)



8 Magnesium reacts with hydrochloric acid.



Which statement about this reaction is correct?

- ☐ A magnesium atoms act as oxidising agents
- ☐ B hydrogen molecules act as reducing agents
- ☐ C hydrogen ions act as oxidising agents
- ☐ D chloride ions act as oxidising agents

(Total for Question 8 = 1 mark)

9 Which of these compounds does **not** produce a colour in a flame test, and produces an alkaline gas when warmed with sodium hydroxide solution?

- ☐ A  $\text{Ca(OH)}_2$
- ☐ B  $\text{Mg(OH)}_2$
- ☐ C  $\text{NH}_4\text{Cl}$
- ☐ D  $\text{BeCl}_2$

(Total for Question 9 = 1 mark)

10 Which of these **increases** as Group 7 is descended?

- ☐ A oxidising ability of the molecular halogens
- ☐ B reducing ability of the halide ions
- ☐ C electrostatic attraction between the nucleus and outer shell of electrons
- ☐ D electronegativity of the halogen atoms

(Total for Question 10 = 1 mark)

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- 11 The table shows the amount of energy released per gram when some alkanes are burned in excess oxygen under standard conditions.

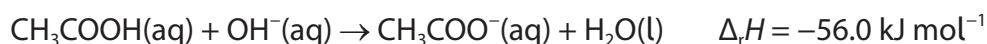
| Alkane  | Energy released / $\text{kJ g}^{-1}$ |
|---------|--------------------------------------|
| methane | 55.6                                 |
| ethane  | 52.0                                 |
| propane | 50.4                                 |
| butane  | 49.6                                 |

Which alkane has a standard enthalpy change of combustion of  $-2877 \text{ kJ mol}^{-1}$ ?

- ☐ A methane
- ☐ B ethane
- ☐ C propane
- ☐ D butane

(Total for Question 11 = 1 mark)

- 12 The enthalpy changes for two reactions are shown.



What is the enthalpy change for the dissociation of  $\text{CH}_3\text{COOH}(\text{aq})$  into  $\text{CH}_3\text{COO}^-(\text{aq})$  and  $\text{H}^+(\text{aq})$  ions, in  $\text{kJ mol}^{-1}$ ?

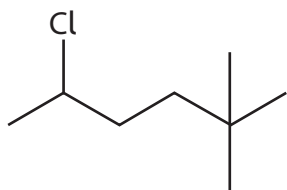
- ☐ A +113.2
- ☐ B +1.2
- ☐ C -1.2
- ☐ D -113.2

(Total for Question 12 = 1 mark)

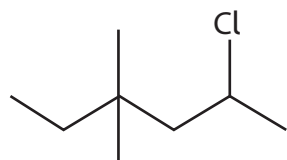


13 What is the skeletal formula of 2-chloro-4,4-dimethylhexane?

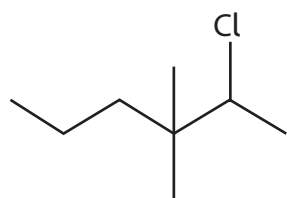
☐ A



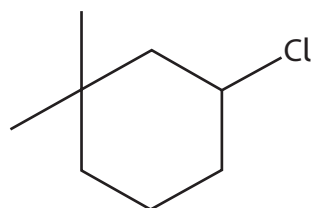
☐ B



☐ C



☐ D



(Total for Question 13 = 1 mark)

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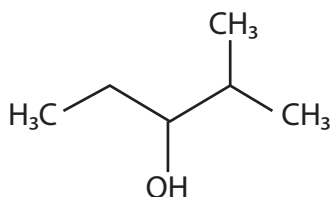


**14** Some alcohols react with concentrated phosphoric(V) acid,  $\text{H}_3\text{PO}_4$ , to form alkenes.

(a) What type of reaction occurs?

- ☐ **A** addition
- ☐ **B** elimination
- ☐ **C** hydrolysis
- ☐ **D** substitution

(b) The structure of 2-methylpentan-3-ol is shown.



This alcohol reacts with concentrated phosphoric(V) acid.

How many different alkenes can form?

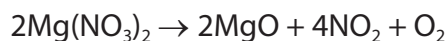
- ☐ **A** one
- ☐ **B** two
- ☐ **C** three
- ☐ **D** four

(Total for Question 14 = 2 marks)

Use this space for any rough working. Anything you write in this space will gain no credit.



15 Magnesium nitrate decomposes on heating.



What is the total volume of gas formed at room temperature and pressure (r.t.p.) when 0.005 mol of magnesium nitrate decomposes completely?

[Molar volume of a gas at r.t.p. =  $24\,000\text{ cm}^3\text{ mol}^{-1}$ ]

- ☐ A  $600\text{ cm}^3$
- ☐ B  $300\text{ cm}^3$
- ☐ C  $240\text{ cm}^3$
- ☐ D  $120\text{ cm}^3$

(Total for Question 15 = 1 mark)

16 How many moles are there in  $15.1\text{ cm}^3$  of liquid propan-1-ol?

[Density of propan-1-ol =  $0.80\text{ g cm}^{-3}$   $M_r$  of propan-1-ol = 60]

- ☐ A  $(0.80 \times 15.1) \div 60$
- ☐ B  $0.80 \div (60 \times 15.1)$
- ☐ C  $60 \div (0.80 \times 15.1)$
- ☐ D  $(60 \times 15.1) \div 0.80$

(Total for Question 16 = 1 mark)

17 The oxidation of propan-1-ol by acidified potassium dichromate(VI) forms propanoic acid with a yield of 36 % by mass.

What mass of propan-1-ol is needed to form 37.0 g of propanoic acid in this reaction?

[ $M_r$  of propanoic acid = 74  $M_r$  of propan-1-ol = 60]

- ☐ A 10.8 g
- ☐ B 36.0 g
- ☐ C 40.8 g
- ☐ D 83.3 g

(Total for Question 17 = 1 mark)



**18** Barium chloride solution,  $\text{BaCl}_2(\text{aq})$ , reacts with gallium sulfate solution,  $\text{Ga}_2(\text{SO}_4)_3(\text{aq})$  to form a precipitate of barium sulfate,  $\text{BaSO}_4(\text{s})$ .

What is the minimum volume of  $0.100 \text{ mol dm}^{-3}$  barium chloride needed to precipitate **all** the sulfate ions in  $200 \text{ cm}^3$  of  $0.05 \text{ mol dm}^{-3}$  gallium sulfate?

- ☐ **A**  $100 \text{ cm}^3$
- ☐ **B**  $200 \text{ cm}^3$
- ☐ **C**  $300 \text{ cm}^3$
- ☐ **D**  $400 \text{ cm}^3$

(Total for Question 18 = 1 mark)

**TOTAL FOR SECTION A = 20 MARKS**



## SECTION B

**Answer ALL the questions. Write your answers in the spaces provided.**

- 19** Ethanol can be made in industry by the reaction of ethene with steam, using a phosphoric(V) acid catalyst.



The reaction is carried out at 300°C and 60 atm. An initial yield of 5 % is achieved when the ethene and steam first pass through the reactor.

- \*(a) Explain the chemical reasons for the conditions used and why such a low initial yield is acceptable in the industrial process.

(6)

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Handwriting practice area with 24 horizontal dotted lines.



P 6 7 7 4 8 A 0 1 3 2 8

(b) The product of the reaction is a mixture of ethanol and water.

Explain why ethanol and water mix together fully.

You may find it helpful to draw a diagram.

(3)

.....

.....

.....

.....

.....

(c) Ethanol can be oxidised using a solution of acidified potassium dichromate(VI),  $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$ .

Ethanoic acid and another organic compound, **Y**, are both possible products.

(i) Draw the structure of **Y**.

(1)

(ii) State the conditions needed to maximise the yield of each product.

(2)

Conditions for maximum yield of **Y**

.....

.....

Conditions for maximum yield of ethanoic acid

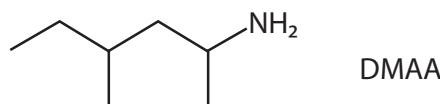
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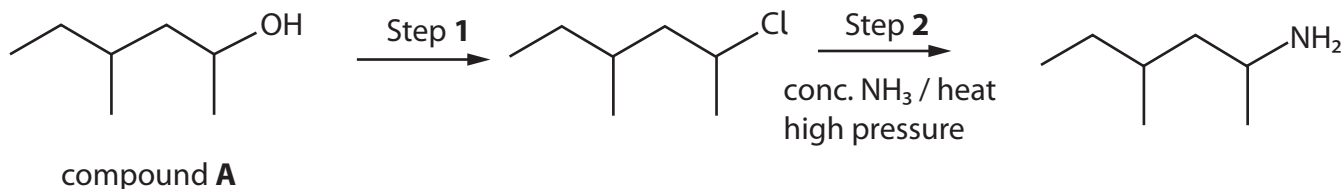
(Total for Question 19 = 12 marks)



20 The compound DMAA was originally synthesised as a decongestant.



A suggested synthetic route for DMAA is shown.



(a) (i) Give the systematic name of compound **A**.

(1)

(ii) Identify, by name or formula, a suitable reagent for Step 1.

(1)

(iii) Give the mechanism for the reaction in Step 2.  
Include curly arrows, and any relevant dipoles and lone pairs.

(4)

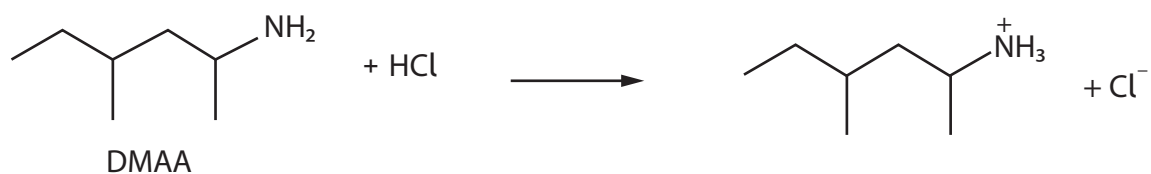
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- (b) DMAA is only slightly soluble in water but dissolves readily in hydrochloric acid to form an aqueous solution.

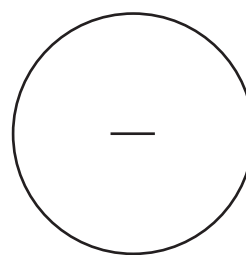
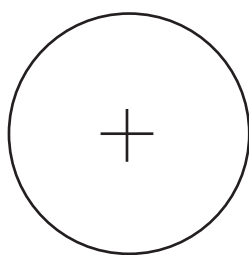


- (i) Explain the type of bonding that occurs between the nitrogen atom and the hydrogen ion, when the positive ion forms.

(3)

- (ii) Complete the diagrams to show how the ions formed in the reaction between DMAA and hydrochloric acid interact with water molecules.

(3)



(Total for Question 20 = 12 marks)





21 This question is about ethanoic acid and some related salts.

- (a) A test to confirm the presence of an aqueous acid is adding a small amount of solid sodium carbonate to the solution.

Describe **two** observations you would **see** in this test.

(2)

.....

.....

.....

.....

- (b) Sodium ethanoate is a component of reusable hand warmers.  
In use, a supersaturated solution of sodium ethanoate recrystallises to form solid hydrated sodium ethanoate, releasing energy.



A hand warmer has a mass of 63.2 g and forms 20.1 g of hydrated sodium ethanoate on recrystallisation.

Calculate the maximum temperature reached by the hand warmer if its initial temperature is 5.0°C.

[Specific heat capacity of the hand warmer = 3.0 J °C<sup>-1</sup> g<sup>-1</sup>]

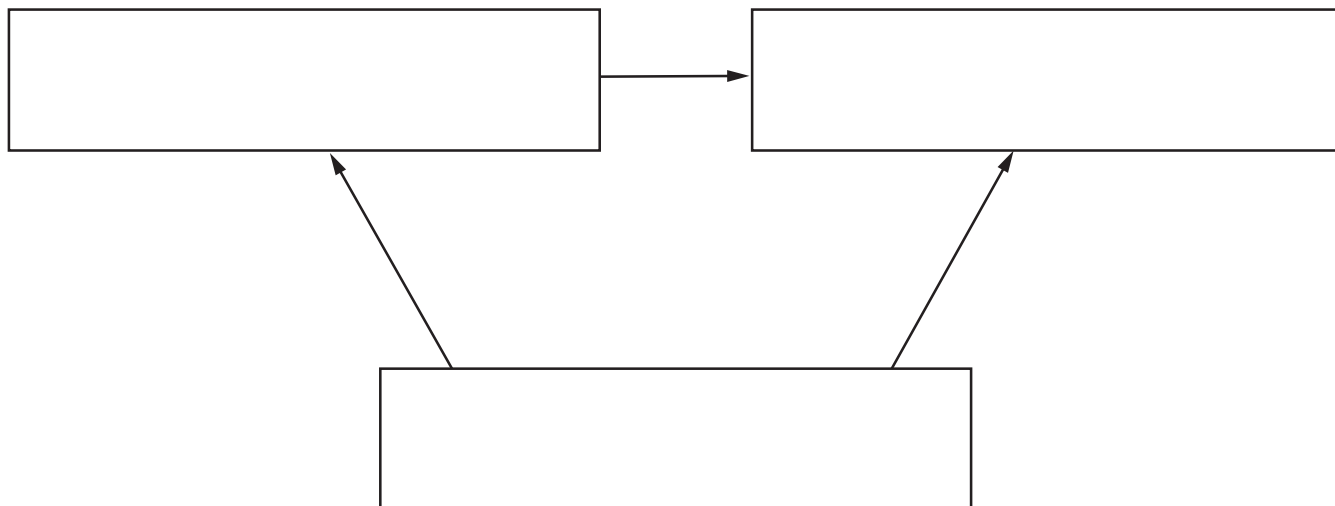
(5)



- (c) Ammonium ethanoate,  $\text{CH}_3\text{COONH}_4(\text{s})$ , is used to control the pH of foods. It can be formed by the reaction of pure ethanoic acid,  $\text{CH}_3\text{COOH}(\text{l})$ , with ammonium carbonate,  $(\text{NH}_4)_2\text{CO}_3(\text{s})$ .

Calculate the standard enthalpy change for this reaction by completing the Hess cycle and using the data shown.

(5)



| Compound                               | Enthalpy change of formation / $\text{kJ mol}^{-1}$ |
|----------------------------------------|-----------------------------------------------------|
| $\text{CH}_3\text{COOH}(\text{l})$     | -484.5                                              |
| $(\text{NH}_4)_2\text{CO}_3(\text{s})$ | -939.9                                              |
| $\text{CH}_3\text{COONH}_4(\text{s})$  | -586.3                                              |
| $\text{CO}_2(\text{g})$                | -393.5                                              |
| $\text{H}_2\text{O}(\text{l})$         | -285.8                                              |



- (d) Ammonium carbonate,  $(\text{NH}_4)_2\text{CO}_3$ , is an ingredient in cleaning solutions for camera lenses.

These are aqueous solutions which contain no more than 1.8 g of ammonium carbonate in  $100 \text{ cm}^3$  of solution.

Calculate the maximum concentration, in  $\text{mol dm}^{-3}$ , of ammonium carbonate in such a solution.

(2)

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**(Total for Question 21 = 14 marks)**

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**TOTAL FOR SECTION B = 38 MARKS**



## SECTION C

Answer ALL the questions. Write your answers in the spaces provided.

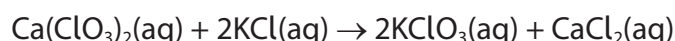
**22** Potassium chlorate(V),  $\text{KClO}_3$ , is a crystalline solid used in fireworks.

It is produced by the Liebig Process in two stages.

Stage **1** Chlorine gas is passed through hot calcium hydroxide solution forming calcium chlorate(V),  $\text{Ca}(\text{ClO}_3)_2$ .



Stage **2** Potassium chloride solution is then added to form potassium chlorate(V).



The solution is heated to reduce its volume and then allowed to crystallise.

The crystals are filtered off.

The remaining filtrate is evaporated further to obtain more crystals.

- (a) (i) Write the **overall** equation for the Liebig Process.  
State symbols are not required.

(1)

- (ii) Calculate the **overall** atom economy by mass for the production of potassium chlorate(V),  $\text{KClO}_3$ , using your equation in (a)(i).

(3)

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- (b) Explain the type of reaction that takes place in Stage 1 of the Liebig Process, using oxidation numbers.



(3)

- (c) The crystals of potassium chlorate(V) formed also contain some halide ion impurities.

- (i) Describe a chemical test on a solution of these crystals to **confirm** that the impurities present are chloride ions rather than bromide ions. Include the expected results.

(3)



- (ii) 1.52 g of impure potassium chlorate(V), formed in the Liebig Process, was heated until the mass of solid remaining was constant at 1.02 g.

The reaction that occurred was



The impurities present did not decompose on heating.

Calculate the percentage purity of the sample.

Give your answer to an appropriate number of significant figures.

(5)

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- (d) In fireworks, potassium chlorate(V) decomposes. This thermal decomposition takes place in two stages with a solid catalyst.



- (i) Give the systematic name of  $\text{KClO}_4$ .

(1)

- (ii) A student investigated the role of the catalyst in this reaction.

**Procedure**

Step 1 Heat a sample of  $\text{KClO}_3$ , in a test tube, with a known mass of insoluble catalyst until the mass remains constant.

Step 2 Mix the contents of the test tube with water.

Step 3 Filter the mixture and rinse with deionised water.

Step 4 Dry the remaining solid.

Step 5 Measure the mass of the dry solid.

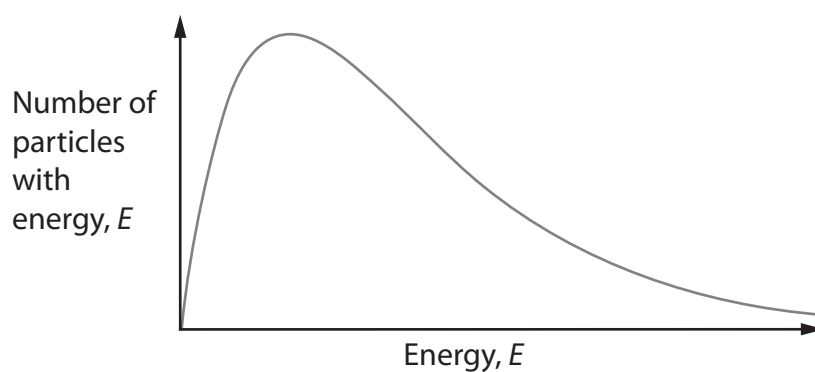
Explain how each of the steps in this procedure is needed to show that the catalyst is **not** used up in this reaction.

(4)



(e) Explain, using the diagram, how a catalyst speeds up a chemical reaction.

(2)



(Total for Question 22 = 22 marks)

**TOTAL FOR SECTION C = 22 MARKS**  
**TOTAL FOR PAPER = 80 MARKS**





# The Periodic Table of Elements

| 1                                                                                                                                                                                                                                                                                                                  | 2                                        | 3                                    | 4                                     | 5                                     | 6                                     | 7                                    | 0 (8)                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|--------------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|--------------------------------------|---------------------------------------|
| 6.9<br><b>Li</b><br>lithium<br>3                                                                                                                                                                                                                                                                                   | 9.0<br><b>Be</b><br>beryllium<br>4       | 10.8<br><b>B</b><br>boron<br>5       | 12.0<br><b>C</b><br>carbon<br>6       | 14.0<br><b>N</b><br>nitrogen<br>7     | 16.0<br><b>O</b><br>oxygen<br>8       | 19.0<br><b>F</b><br>fluorine<br>9    | 20.2<br><b>Ne</b><br>neon<br>10       |
| 23.0<br><b>Na</b><br>sodium<br>11                                                                                                                                                                                                                                                                                  | 24.3<br><b>Mg</b><br>magnesium<br>12     | 27.0<br><b>Al</b><br>aluminium<br>13 | 28.1<br><b>Si</b><br>silicon<br>14    | 31.0<br><b>P</b><br>phosphorus<br>15  | 32.1<br><b>S</b><br>sulfur<br>16      | 35.5<br><b>Cl</b><br>chlorine<br>17  | 39.9<br><b>Ar</b><br>argon<br>18      |
| 39.1<br><b>K</b><br>potassium<br>19                                                                                                                                                                                                                                                                                | 40.1<br><b>Ca</b><br>calcium<br>20       | 69.7<br><b>Ga</b><br>gallium<br>31   | 72.6<br><b>Ge</b><br>germanium<br>32  | 74.9<br><b>As</b><br>arsenic<br>33    | 79.0<br><b>Se</b><br>selenium<br>34   | 79.9<br><b>Br</b><br>bromine<br>35   | 83.8<br><b>Kr</b><br>krypton<br>36    |
| 85.5<br><b>Rb</b><br>rubidium<br>37                                                                                                                                                                                                                                                                                | 87.6<br><b>Sr</b><br>strontium<br>38     | 114.8<br><b>In</b><br>indium<br>49   | 118.7<br><b>Sn</b><br>tin<br>50       | 121.8<br><b>Sb</b><br>antimony<br>51  | 127.6<br><b>Te</b><br>tellurium<br>52 | 126.9<br><b>I</b><br>iodine<br>53    | 131.3<br><b>Xe</b><br>xenon<br>54     |
| 132.9<br><b>Cs</b><br>caesium<br>55                                                                                                                                                                                                                                                                                | 137.3<br><b>Ba</b><br>barium<br>56       | 204.4<br><b>Tl</b><br>thallium<br>81 | 207.2<br><b>Pb</b><br>lead<br>82      | 209.0<br><b>Bi</b><br>bismuth<br>83   | [209]<br><b>Po</b><br>polonium<br>84  | [210]<br><b>At</b><br>astatine<br>85 | [222]<br><b>Rn</b><br>radon<br>86     |
| [223]<br><b>Fr</b><br>francium<br>87                                                                                                                                                                                                                                                                               | [226]<br><b>Ra</b><br>radium<br>88       | 200.6<br><b>Hg</b><br>mercury<br>80  | 207.2<br><b>Pb</b><br>lead<br>82      | 209.0<br><b>Bi</b><br>bismuth<br>83   | [209]<br><b>Po</b><br>polonium<br>84  | [210]<br><b>At</b><br>astatine<br>85 | [222]<br><b>Rn</b><br>radon<br>86     |
| Elements with atomic numbers 112-116 have been reported but not fully authenticated                                                                                                                                                                                                                                |                                          |                                      |                                       |                                       |                                       |                                      |                                       |
| <div> <div> <div>1.0<br/><b>H</b><br/>hydrogen<br/>1</div> <div>relative atomic mass<br/>atomic symbol<br/>name<br/>atomic (proton) number</div> </div> <div> <div>Key</div> <div> <div>relative atomic mass</div> <div>atomic symbol</div> <div>name</div> <div>atomic (proton) number</div> </div> </div> </div> |                                          |                                      |                                       |                                       |                                       |                                      |                                       |
| 140<br><b>Ce</b><br>cerium<br>58                                                                                                                                                                                                                                                                                   | 141<br><b>Pr</b><br>praseodymium<br>59   | 144<br><b>Nd</b><br>neodymium<br>60  | 147<br><b>Pm</b><br>promethium<br>61  | 150<br><b>Sm</b><br>samarium<br>62    | 152<br><b>Eu</b><br>europium<br>63    | 157<br><b>Gd</b><br>gadolinium<br>64 | 159<br><b>Tb</b><br>terbium<br>65     |
| 232<br><b>Th</b><br>thorium<br>90                                                                                                                                                                                                                                                                                  | [231]<br><b>Pa</b><br>protactinium<br>91 | 238<br><b>U</b><br>uranium<br>92     | [237]<br><b>Np</b><br>neptunium<br>93 | [242]<br><b>Pu</b><br>plutonium<br>94 | [243]<br><b>Am</b><br>americium<br>95 | [247]<br><b>Cm</b><br>curium<br>96   | [245]<br><b>Bk</b><br>berkelium<br>97 |
| <div> <div>* Lanthanide series</div> <div>* Actinide series</div> </div>                                                                                                                                                                                                                                           |                                          |                                      |                                       |                                       |                                       |                                      |                                       |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163<br><b>Dy</b><br>dysprosium<br>66                                                                                                                                                                                                                                                                               | 165<br><b>Ho</b><br>holmium<br>67        | 167<br><b>Er</b><br>erbium<br>68     | 169<br><b>Tm</b><br>thulium<br>69     | 173<br><b>Yb</b><br>ytterbium<br>70   | 175<br><b>Lu</b><br>lutetium<br>71    | [253]<br><b>Fm</b><br>fermium<br>100 | [254]<br><b>No</b><br>nobelium<br>102 |
| 163                                                                                                                                                                                                                                                                                                                |                                          |                                      |                                       |                                       |                                       |                                      |                                       |